LAB WEEK 4

SQL Practice Queries

The Pressly Ad Agency Database

Practice Query 1

Display the names of all campaign managers.

SQL query

```
SELECT CampaignMgrName
FROM CampaignManager;
```

Expected result

CampaignMgrName
Roberta
Sue
John
Nancy

Practice Query 2

Display all campaigns managed by Roberta.

SQL query

```
SELECT ac.*
FROM AdCampaign ac
JOIN CampaignManager cm
    ON ac.CampaignMgrID = cm.CampaignMgrID
WHERE cm.CampaignMgrName = 'Roberta';
```

Expected result

AdCampaignID	AdCampaignName	StartDate	Duration	CampaignMgrID
111	SummerFun20	2020-06-06	12	CM100
555	AutmnColors20	2020-09-06	3	CM100

Practice Query 3

Display all advertising modes that use Radio.

SQL query

```
SELECT *
FROM Modus
WHERE Media = 'Radio';
```

Expected result

ModusID	Media	Range
3	Radio	Local
4	Radio	National

Practice Query 4

Display the campaign name, media, range and budget percentage for every campaign.

SQL query

```
SELECT
    ac.AdCampaignName,
    m.Media,
    m.`Range`,
    acm.BudgetPctg
FROM AdCampaign ac
JOIN AdCampaignMix acm
    ON ac.AdCampaignID = acm.AdCampaignID
JOIN Modus m
    ON acm.ModusID = m.ModusID;
```

Expected result

AdCampaignName	Media	Range	BudgetPctg
SummerFun20	TV	Local	50
SummerFun20	TV	National	50
SummerZing20	TV	Local	60
SummerZing20	Radio	Local	30
SummerZing20	Print	Local	10
FallBall20	Radio	Local	80
FallBall20	Radio	National	20
AutumnStyle20	Print	National	100
AutmnColors20	Radio	Local	100

Practice Query 5

Display only campaigns that use a National advertising modus.

SQL query

```
SELECT DISTINCT
    ac.AdCampaignID,
    ac.AdCampaignName
FROM AdCampaign ac
JOIN AdCampaignMix acm
    ON ac.AdCampaignID = acm.AdCampaignID
JOIN Modus m
    ON acm.ModusID = m.ModusID
WHERE m.`Range` = 'National';
```

Expected result

AdCampaignID	AdCampaignName
111	SummerFun20
333	FallBall20
444	AutumnStyle20

Note: DISTINCT is used so that a campaign appears only once even if more than one matching row could exist.

Practice Query 6

Display all advertising modes used by SummerZing20.

SQL query

```
SELECT
    m.ModusID,
    m.Media,
    m.`Range`
FROM AdCampaign ac
JOIN AdCampaignMix acm
    ON ac.AdCampaignID = acm.AdCampaignID
JOIN Modus m
    ON acm.ModusID = m.ModusID
WHERE ac.AdCampaignName = 'SummerZing20';
```

Expected result

ModusID	Media	Range
1	TV	Local
3	Radio	Local
5	Print	Local

Practice Query 7

Display each campaign together with its campaign manager.

SQL query

```
SELECT
    ac.AdCampaignID,
    ac.AdCampaignName,
    cm.CampaignMgrName
FROM AdCampaign ac
JOIN CampaignManager cm
    ON ac.CampaignMgrID = cm.CampaignMgrID;
```

Expected result

AdCampaignID	AdCampaignName	CampaignMgrName
111	SummerFun20	Roberta
222	SummerZing20	Sue
333	FallBall20	John
444	AutumnStyle20	Nancy
555	AutmnColors20	Roberta

Practice Query 8

Display campaign/modus combinations where more than 50% of the campaign budget is allocated to one modus.

SQL query

```
SELECT
    ac.AdCampaignName,
    m.Media,
    m.`Range`,
    acm.BudgetPctg
FROM AdCampaign ac
JOIN AdCampaignMix acm
    ON ac.AdCampaignID = acm.AdCampaignID
JOIN Modus m
    ON acm.ModusID = m.ModusID
WHERE acm.BudgetPctg > 50;
```

Expected result

AdCampaignName	Media	Range	BudgetPctg
SummerZing20	TV	Local	60
FallBall20	Radio	Local	80
AutumnStyle20	Print	National	100
AutmnColors20	Radio	Local	100

Practice Query 9

Count the number of advertising modes used by each campaign.

SQL query

```
SELECT
    ac.AdCampaignID,
    ac.AdCampaignName,
    COUNT(acm.ModusID) AS NumberOfModes
FROM AdCampaign ac
JOIN AdCampaignMix acm
    ON ac.AdCampaignID = acm.AdCampaignID
GROUP BY
    ac.AdCampaignID,
    ac.AdCampaignName;
```

Expected result

AdCampaignID	AdCampaignName	NumberOfModes
111	SummerFun20	2
222	SummerZing20	3
333	FallBall20	2
444	AutumnStyle20	1
555	AutmnColors20	1

Note: COUNT() counts the number of modus records associated with each campaign. GROUP BY creates one result row per campaign.

Practice Query 10

Calculate the total budget percentage allocated to each campaign.

SQL query

```
SELECT
    ac.AdCampaignID,
    ac.AdCampaignName,
    SUM(acm.BudgetPctg) AS TotalBudgetPctg
FROM AdCampaign ac
JOIN AdCampaignMix acm
    ON ac.AdCampaignID = acm.AdCampaignID
GROUP BY
    ac.AdCampaignID,
    ac.AdCampaignName;
```

Expected result

AdCampaignID	AdCampaignName	TotalBudgetPctg
111	SummerFun20	100
222	SummerZing20	100
333	FallBall20	100
444	AutumnStyle20	100
555	AutmnColors20	100

Note: A total of 100 for every campaign confirms that the complete campaign budget has been allocated across its advertising modes.

Quick Check

All five campaigns have a total budget allocation of 100%, and the queries demonstrate selection, filtering, joins, DISTINCT, COUNT(), SUM(), and GROUP BY using the normalized database.